BCP and Countable Capture Are Exact Opposites

Run `fa_banach_001`, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^*-algebra $C(K)$ has BCP/UBCP exactly when K has a countable π-basis, it asks for a noncommutative version:

> $\{U_\alpha\}_{\alpha \in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Let X be a nonzero normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. We say that X has *countable capture* if for every countable $C \subset X$ there is $u \in S_X$ such that

$$\operatorname{dist}(c, \mathbb{F}u) = \|c\| \qquad (c \in C).$$

This is equivalent to the quotient-map and common-kernel formulations in the earlier packet

`2311.14257_countable_capture_characterization`.

We use the standard BCP form: X has BCP if and only if there are countably many centres $x_n \in X \setminus \{0\}$ such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|),$$

where the balls are open. Indeed, any ball not containing the origin has radius strictly smaller than the norm of its centre, and enlarging its radius to the centre norm gives the displayed form. Conversely, $0 \notin B(x_n, \|x_n\|)$ because the balls are open.

Main Theorem

Theorem 1 (exact BCP/capture equivalence)**.** *For every nonzero normed space X, the following are equivalent:*

1. *X fails BCP;*

2. *X has countable capture.*

Equivalently, X has BCP if and only if there is a countable set $C \subset X$ such that for every $u \in S_X$ there exists $c \in C$ with

$$\operatorname{dist}(c, \mathbb{F}u) < \|c\|.$$

Corollary 1 (C*-algebras)**.** *For every C^*-algebra A, the underlying Banach space of A has BCP if and only if A does not have countable capture. Hence A fails BCP if and only if it has countable capture.*

Idea of the Proof

The one-way obstruction was already known in the run: if every countable family of proposed centres admits a unit vector whose line preserves all their distances, then that unit vector escapes every BCP ball.

The missing reverse implication is a scalar-rescaling trick. If a fixed countable set C blocks countable capture, then for every $u \in S_X$ some $c \in C$ satisfies

$$\operatorname{dist}(c, \mathbb{F}u) < \|c\|.$$

Thus c is closer than $\|c\|$ to some scalar multiple λu. The scalar λ is nonzero, and after dividing by it we get

$$\|u - \lambda^{-1}c\| < \|\lambda^{-1}c\|.$$

So u lies in a standard BCP ball centred at a scalar multiple of c. A countable dense set of scalars keeps the family of such multiples countable.

Proof

Proof of Theorem 1. First suppose X has countable capture. Let $(x_n) \subset X$ be any countable family of proposed BCP centres. By countable capture, choose $u \in S_X$ such that

$$\operatorname{dist}(x_n, \mathbb{F}u) = \|x_n\| \qquad (n \geq 1).$$

Then, for every n,

$$\|u - x_n\| \geq \operatorname{dist}(x_n, \mathbb{F}u) = \|x_n\|.$$

Hence $u \notin B(x_n, \|x_n\|)$ for every n. Since the centre family was arbitrary, no countable BCP cover exists. Thus X fails BCP.

Conversely, suppose X does not have countable capture. Then there is a countable set $C \subset X$ such that for every $u \in S_X$ there exists $c \in C$ with

$$\operatorname{dist}(c, \mathbb{F}u) < \|c\|.$$

The element c cannot be zero. Therefore for some scalar $\lambda \in \mathbb{F}$ we have

$$\|c - \lambda u\| < \|c\|.$$

The scalar λ cannot be zero, since otherwise the inequality would read $\|c\| < \|c\|$. Put $\mu = \lambda^{-1}$. Dividing by $|\lambda|$ gives

$$\|u - \mu c\| < \|\mu c\|.$$

Let $D \subset \mathbb{F} \setminus \{0\}$ be a countable dense set, for instance $\mathbb{Q} \setminus \{0\}$ in the real case and $(\mathbb{Q} + i\mathbb{Q}) \setminus \{0\}$ in the complex case. For fixed u and c, the set

$$\{\alpha \in \mathbb{F} : \|u - \alpha c\| < \|\alpha c\|\}$$

is open in $\mathbb{F}$ and contains μ. Choose $q \in D$ in this open set. Then

$$\|u - qc\| < \|qc\|.$$

Thus every $u \in S_X$ belongs to one of the balls

$$B(qc, \|qc\|), \qquad c \in C, \ q \in D.$$

The set of such centres is countable, and all displayed open balls omit the origin. Hence X has BCP.

This proves both the equivalence X fails BCP if and only if X has countable capture and the displayed equivalent positive formulation. □

Proof of Corollary 1. Apply Theorem 1 to the underlying complex Banach space of the C^*-algebra A. □

Consequences for the C*-Algebra Program

The theorem proves the lane-19 conjecture

$$C^*\text{-algebra BCP failure} \iff \text{countable capture.}$$

In fact, no C^*-specific structure is needed for this equivalence.

The many previous negative packets now have a common interpretation: they produce countable-capture witnesses. The positive packets produce explicit countable sets that block capture, often with uniform constants. For example, the countable product theorem gives UBCP by constructing a countable family of centres c for which every unit vector satisfies a strict inequality $\|u - c\| < \|c\|$.

Limitations

This packet settles the exact BCP/capture equivalence. It does not by itself settle the stronger quantitative question of UBCP. A parallel quantitative version would need uniform control of the gap between a covering radius and the norm of the centre, plus boundedness of the radii. The earlier C^*-algebra positive packets provide UBCP in their model classes, but this theorem does not prove that every C^*-algebra with BCP has UBCP.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, BCP, UBCP, countable capture, quotient-line characterization, Birkhoff–James orthogonality, and the earlier lane-19 capture packets. The previous packet

`2311.14257_countable_capture_characterization`

proved the quotient and dual forms of countable capture, while

`2311.14257_countable_capture_bcp_obstruction`

proved only the forward implication "countable capture implies failure of BCP." The reverse implication in this packet is the new scalar-rescaling step. Web searches during the surrounding June 17, 2026 pushes found the source BCP literature and adjacent operator-space papers, but no prior record of this exact capture/BCP equivalence.

Human Review Focus

Reviewers should check:

1. the negation of countable capture;
2. the fact that $\lambda \neq 0$ in the strict distance inequality;
3. the passage from $\|c - \lambda u\| < \|c\|$ to $\|u - \lambda^{-1}c\| < \|\lambda^{-1}c\|$;
4. the use of a countable dense scalar set to keep the final BCP centres countable;
5. the stated limitation that this is an exact BCP theorem, not a UBCP classification.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Run `fa_banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.

[3] Run `fa_banach_001`, lane 19, *A Countable-Capture Obstruction for the Ball-Covering Property*, solution packet `2311.14257_countable_capture_bcp_obstruction`, 2026.